Data Analysis Project with Pandas, NumPy, and Matplotlib

Total Marks: 40

Choose a real dataset, answer meaningful questions, justify your data decisions, and present evidence-based findings.

Project goal: carry out a mini data investigation using Pandas, NumPy, and Matplotlib. Your work must show data understanding, justified cleaning, feature engineering, analytical depth, clear visual communication, and honest reflection on limitations.

1. Project Overview

This project is designed to test not only whether you can use Python libraries, but whether you can think like an analyst. A strong submission will define clear questions, prepare the data carefully, perform meaningful analysis, and defend its conclusions with evidence.

- This is not a chart-making task. Random plots without a purpose will not score highly.
- You must answer 2 to 3 meaningful questions using data, not just describe the dataset.
- Your final work should demonstrate technical skill, reasoning, interpretation, and professionalism.

2. Required Tools

- Pandas: for loading, inspecting, cleaning, filtering, grouping, aggregating, transforming, and summarizing the dataset.
- NumPy: for meaningful numerical work such as custom calculations, array-based processing, percentiles, z-scores, normalization, or other statistical computation.
- Matplotlib: for visualizations that directly support your questions and findings.
- You may work in Google Colab, Jupyter Notebook, or a local Python environment.
- You may use small supporting libraries if necessary, but the core analysis must clearly depend on Pandas, NumPy, and Matplotlib.
- It is not acceptable to import one of the required libraries without using it meaningfully.

3. Dataset Requirements

- Your dataset must be real and obtained from a valid source such as Kaggle, UCI Machine Learning Repository, data.gov, the World Bank, WHO, UNESCO, or another credible open-data platform.
- It must contain at least 1000 rows and at least 6 columns.
- It must be rich enough to support comparison, interpretation, and multiple forms of analysis.
- You must mention the dataset source clearly in both your code/notebook and your report.
- Do not choose a dataset that is tiny, trivial, mostly empty, or unsuitable for meaningful analysis.
- Strong preference: choose a dataset that includes mixed data types, missing values, date/time information, categorical and numerical variables, or enough complexity to make your analysis interesting.

Examples of suitable themes: student performance, sales, finance, transport, health, sports, environment, public policy, business operations, or survey data.

4. What Your Project Must Include

4.1 Problem Definition

- State what dataset you selected.
- Explain what real topic or issue the dataset represents.
- Write 2 to 3 meaningful analytical questions that your project will answer.
- Briefly explain why those questions matter.

4.2 Data Understanding

- Show the dataset shape, column names, data types, sample rows, summary statistics, and missing-value summary.
- Do not only print outputs. Briefly explain what those outputs tell you about the structure and quality of the data.

4.3 Data Cleaning and Preparation

- Perform at least 3 meaningful cleaning or preprocessing steps.

- Examples include handling missing values, removing duplicates, converting data types, fixing inconsistent entries, formatting dates, renaming columns, and removing or analyzing outliers.
- For every major cleaning step, write a short justification: what the problem was, what you did, and why that decision was reasonable.

4.4 Feature Engineering

- Create at least 2 new derived columns from existing data.
- Examples include age groups, month or year from a date column, family size, profit margin, normalized values, fare category, rolling average, or a custom score.
- You are not allowed to rely only on the original columns. Your new features should help support later analysis.

4.5 Data Analysis

- Perform at least 6 meaningful analysis operations.
- Your project must include at least 2 subgroup comparisons.
- It must include at least 1 relationship analysis between variables.
- It must include at least 1 outlier, anomaly, or unusual-pattern analysis.
- It must include at least 1 custom NumPy-based computation.
- Examples of strong analysis include filtering, grouping and aggregation, ranking, trend analysis, distribution analysis, correlation-style comparison, frequency analysis, and custom metrics.

4.6 Visualization

- Create at least 4 meaningful charts using Matplotlib.
- Every chart must have a clear title, labeled axes where appropriate, and readable formatting.
- For each chart, write 2 to 3 sentences explaining what it shows, why it matters, and what insight can be taken from it.
- Accepted chart types include line charts, bar charts, histograms, scatter plots, box plots, pie charts where appropriate, heatmaps built with Matplotlib, and multi-panel figures.

4.7 Findings and Discussion

- Write at least 5 clear, evidence-based insights from your analysis.
- Avoid weak statements such as “the graph looks good” or “the data is interesting.”
- Strong findings should connect your evidence back to the questions you originally asked.

4.8 Limitations

- Include a short section on limitations.
- Discuss issues such as missing values, sample bias, incomplete variables, preprocessing choices, or the fact that correlation does not prove causation.

5. Code Submission Instructions

- Your project code must be submitted through either a GitHub repository link or a Google Colab notebook link.
- You must upload the code link to the Google Sheet provided by the instructor.
- Your code should be clearly organized into sections such as: import libraries, load data, inspect data, clean data, engineer features, analyze, visualize, and summarize findings.
- Your notebook or repository must clearly include your name, student ID, project title, and dataset source.
- Your code/notebook must run correctly. Broken links, inaccessible repositories, or missing files may lead to mark deduction.
- If you use GitHub, make sure the repository is public or accessible to the instructor.
- If you use Google Colab, make sure the sharing permission allows the instructor to view the notebook.

6. How to Submit

1. Open the Google Sheet provided by the me.
2. Enter your required information in the assigned row, including your name, student ID, and the GitHub or Google Colab link to your project.
3. Check the link yourself before submission to make sure it opens properly and can be viewed.
4. Upload your project report separately using the submission method announced by me.
5. Submit before the deadline announced in class.

Important: A project cannot be evaluated properly if the code link does not work or if the report is missing.

7. Project Report Requirements

Recommended length: 3 to 5 pages.

- Project title
- Student name and student ID
- Dataset name and source link
- Project objective and analytical questions
- Dataset description
- Data cleaning and preparation
- Feature engineering
- Analysis and visualizations
- Key findings
- Limitations
- Conclusion

Requirement: Include selected screenshots or exported figures from your analysis to make the report easier to understand.

8. Presentation / Viva Expectation

- You may be asked to explain your work orally.
- Be prepared to answer questions such as: Why did you choose this dataset? Why did you clean the data in that way? Why did you create those new features? Which result do you trust the most? What is one weakness of your analysis?
- If you cannot explain your own project clearly, marks may be reduced.

9. Marking Scheme (40 Marks)

Component	Marks	What is expected
Problem definition and dataset selection	5	Clear project objective, appropriate dataset, valid source, and meaningful analytical questions.
Data understanding and cleaning	8	Proper inspection of the dataset, identification of issues, meaningful cleaning steps, and justification of cleaning decisions.
Feature engineering	5	At least 2 useful derived features that support later analysis and are logically motivated.
Data analysis using Pandas and NumPy	10	Analytical depth, subgroup comparisons, relationship analysis, outlier or anomaly analysis, and meaningful NumPy usage beyond a trivial import.
Visualization with Matplotlib	6	At least 4 useful charts with good labels, readable formatting, and clear interpretation.
Findings, limitations, and conclusion	4	Evidence-based insights, honest discussion of limitations, and a logical conclusion.
Code quality, report quality, and submission completeness	2	Readable code, organized report, working submission link, and complete submission materials.

10. Academic Integrity and Project Quality

- Your work must be your own. Copying notebooks, repositories, or reports from the internet or from classmates is not acceptable.
- You may read documentation, tutorials, and examples, but your final submission must show your own understanding and original analysis.
- You must cite the dataset source and any major external reference you relied on.
- Submitting code that you cannot explain may result in mark reduction during presentation or viva.
- Superficial projects that only show basic commands without real analytical depth may receive low marks even if they technically use the required libraries.

11. Final Checklist Before Submission

- ☐ I selected a suitable dataset with at least 1000 rows and 6 columns.
- ☐ I used Pandas, NumPy, and Matplotlib meaningfully.
- ☐ I included at least 3 cleaning or preprocessing steps with justification.
- ☐ I created at least 2 engineered features.
- ☐ I performed at least 6 meaningful analysis operations.
- ☐ I included at least 2 subgroup comparisons.
- ☐ I included at least 1 relationship analysis and 1 outlier or unusual-pattern analysis.
- ☐ I created at least 4 useful charts and interpreted each of them.
- ☐ I wrote at least 5 clear findings and included a limitations section.

- ☐ My GitHub or Google Colab link works properly and has been added to the Google Sheet.
- ☐ My report is complete and ready to submit.
- ☐ My name, student ID, project title, and dataset source are clearly mentioned.

Best practice: choose a dataset that genuinely interests you. Strong questions usually lead to stronger analysis.